

Sri Lanka Institute of Information Technology



IT1040 - Fundamentals of Computing

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Automated Pet Feeder Proposal Document

Group 019

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1. Background

Nowadays there are so many people who are pet owners but are also busy with their own lives. Whether it is professionally, travelling for work or vacations and having unexpected changes in their day-to-day routine, like emergencies or formal events. In such situations, pet owners find it hard to feed their pets at appropriate times. How can they feed their pets on time, and ease their mind to be worry free?

Due to the above-mentioned problems of pets missing meals or being fed at unhealthy times, which may affect their health. The idea of this project is to create a simple but yet intricate system, called an Automated Pet Feeding System, to automatically feed pets at the right times at which was predefined by the user. This system will be helpful for pets so that they are taken care of even when their owners are not at home.

2. Problem and Motivation

2.1 Problem

Pet owners tend to forget or cannot feed their pets on time. This causes underfeeding, overfeeding or irregular feeding, which is not healthy for pets. When the owners are not at home for long periods, manual feeding will be an issue for the pets and owners too.

2.2 Motivation

This Automated Pet Feeder system aims to make day-to-day life easier for both pets and their owners. Ensure that pets get their food on time every day and at the required quantities predefined. When the pet owner knows that their pets will be fed properly at the right time without any doubt, that gives peace of mind to the owners. With sensors, circuits and programming, we will make a reliable system that helps families become more considerate of their pets.

3. Aim and Objectives

3.1 Aim

To build and design automated, an intelligent system that precisely feeds pets at pre-scheduled times, with the help of basic sensors, motors and time tracking algorithms.

3.2 Objectives

- Create a programmable system using an Arduino [1] micro-controller.
- Design an application or a suitable website with an ESP32 [2] module.
- Track feeding times accurately with a Real-Time Clock module (RTC) [3].
- Design a food dispensing mechanism controlled by a Servo Motor [4].
- Implement Ultrasonic Range sensor [5] to track food level.
- Use an LCD display screen [6] to show the levels of food remaining in the container.
- Track food output with Load Cell [7] weighing sensor.
- Evaluate the created automated system for functionality, safety and reliability.
- Ensure that the finalized system is easy to navigate and cost-effective.

4. System Diagram

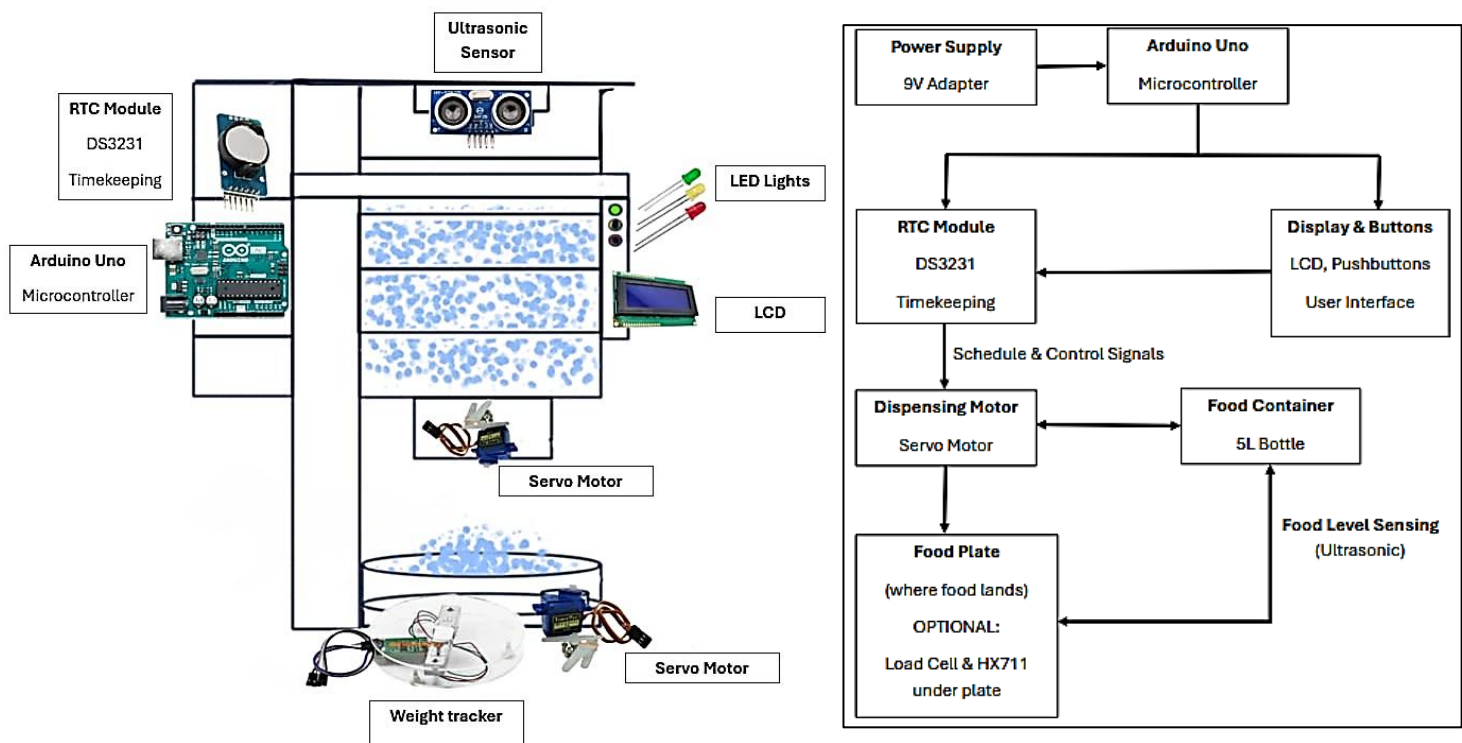


Fig.10. APFie System Diagram. Source: LAS Wijesinghe (Component images adapted from duino.lk [1],[2],[3],[4],[5],[6],[7])

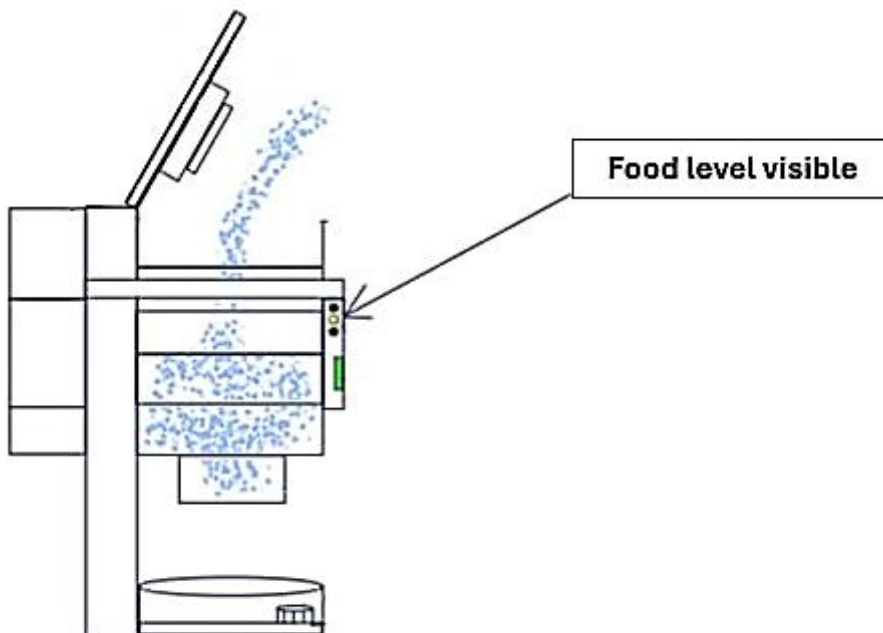


Fig.11. APFie System Diagram. Source: LAS Wijesinghe

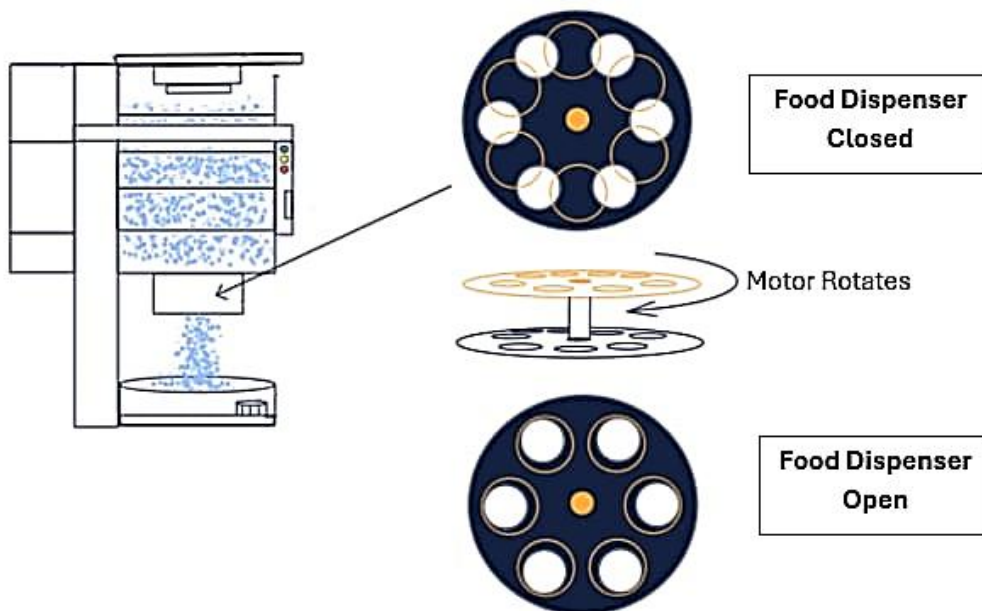


Fig.12. APFie System Diagram. Source: LAS Wijesinghe

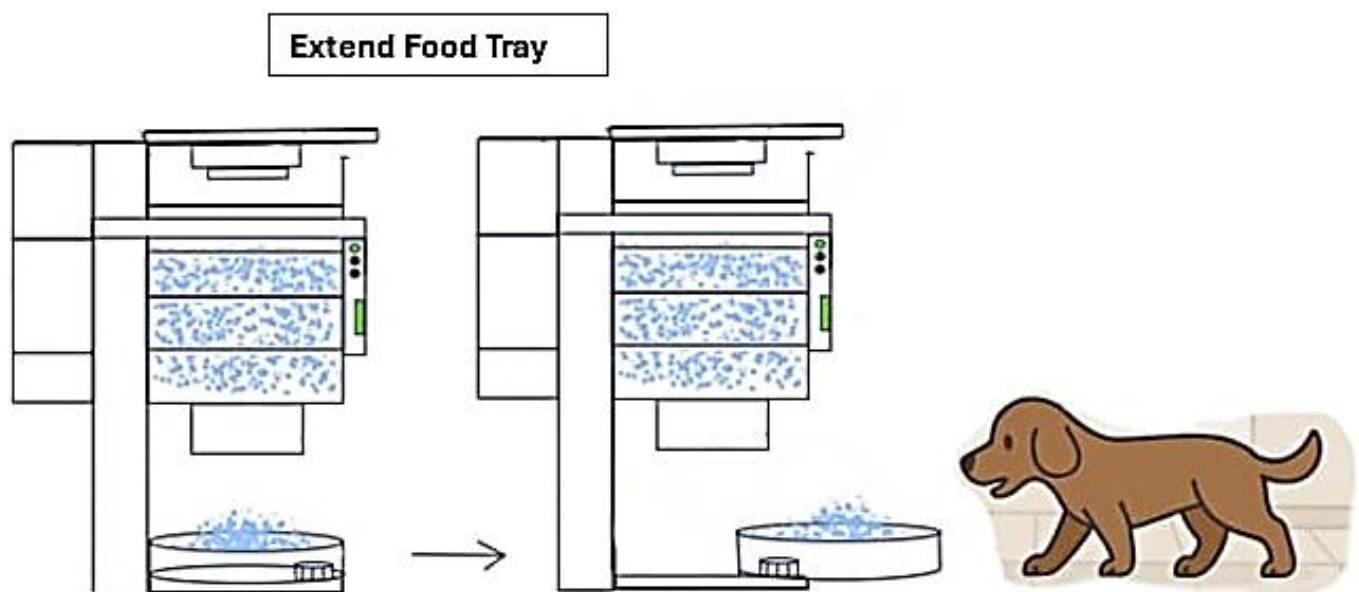


Fig.13. APFie System Diagram. Source: LAS Wijesinghe

5. Methodology

To implement this idea in our everyday life, this project will follow a practical approach that combines both hardware and software components as follows:

- Hardware Tools

- ❖ Arduino Uno (microcontroller)
 - A small device to control the system.
- ❖ ESP32 module
- ❖ Food dispenser
- ❖ Ultra-sonic sensor
 - To check the food levels in the container.
- ❖ Food container with a covered casing
 - To store pet food.
- ❖ Timer (clock module – RTC)
 - To keep track of time for feeding.
- ❖ Servo motor
 - To open the food container and let the food fall into the bowl.
- ❖ Push buttons and LCD display for user input
 - To allow users to set feeding times.

- Software Tools

- ❖ Arduino IDE [8]
- ❖ Tinkercad [9]

6. Evaluation

The system will be evaluated by testing the following things:

- The automated system feeds the pet at the right time.
- Performs accordingly, without any issue over several days and weeks.
- The food portions that are dispensed are precise and consistent.
- It is user friendly and easy to set-up.
- All the electronic and non-electronic parts are safe for use around pets.
- The food dispenser opens and closes at the correct angle of rotation.
- LED lights correctly calibrated with the Ultrasonic Sensor.
- Tray extends only after the food dispenser is closed.
- Structure supports the weight of the 5-liter bottle after food insertion.
- Correct current applied for each sensor and component.
- Solution to prevent blockage of the food dispenser works.
(motor rotates anticlockwise)
- Test whether components are safe from external factors (e.g. water, shaking, or dropping).

7. References

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